University of Science and Technology of Southern Philippines
College of Engineering and Architecture
Bachelor of Science in Computer Engineering

Concept Paper Presentation

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Category: Internet of Things

Proposed Title:
A Modular IoT-Based Digital Twin for Campus-Scale Flood Risk Monitoring and Prediction

Introduction:
Flooding is a recurring environmental issue that affects urban and institutional areas, often caused by intense rainfall, inadequate drainage systems, and rapid land-use changes. Educational campuses, such as the University of Science and Technology of Southern Philippines – Cagayan de Oro (USTP-CDO), are not exempt from these risks, as localized flooding can disrupt mobility, damage infrastructure, and pose safety concerns to students and staff.

Traditional flood assessment methods commonly rely on historical records and static maps, which do not sufficiently capture how rainfall interacts with drainage systems over time. Recent advancements in digital twin technology offer a new approach by enabling the creation of virtual representations of real-world environments that can simulate physical processes under controlled conditions.

Preliminary Literature Review:
Previous studies have shown that digital twin and IoT-based systems are effective for flood analysis, especially in large urban areas. They often combine real-time sensors, hydrodynamic models, and machine learning to predict floods or provide early alerts. While these approaches work well at city or watershed scales, they require complex infrastructure and are expensive to deploy.
In contrast, our study focuses on a campus-scale, modular digital twin that uses tipping rain gauge data and a 2D model of the USTP-CDO campus to simulate rainfall and assess flood-prone areas. This approach is simpler, cost-effective, and tailored for planning and risk assessment rather than real-time monitoring.

2D Digital Twin & Pluvial Flood Modeling

No.	Approach of the Study	Gap Identified	How Our Study Fills the Gap
1	Systematic review of digital twin applications in flood risk using remote sensing & IoT	Focuses on urban scale with complex data needs and high computational demand	Our work scales digital twin to a campus 2D model with simpler data inputs
2	Enhances 1D–2D flood models using high-resolution DEM inputs for better simulation accuracy	Assumes high-resolution spatial data not always available	Our study uses locally measured rainfall + campus layout rather than demanding high-res terrain

IoT-Based Flood Monitoring & Flood Level Classification

No.	Approach of the Study	Gap Identified	How Our Study Fills the Gap
1	IoT pressure sensors monitor flood levels with LoRaWAN transmission	Focused on water level sensing/real-time alerts rather than flood risk prediction	We focus on rainfall-driven simulation + planning, tailored to campus
2	Classifies flood hazard levels by maximum water depth, drainage efficiency, and event frequency.	Requires dense sensor arrays; multi-indicator complexity not tied to simple rainfall thresholds.	We link rainfall intensity thresholds to expected surface water accumulation using a 2D digital twin, suitable for limited infrastructure

Statement of the Problem:

Pluvial flooding has become a recurring issue within the USTP-CDO campus, particularly during periods of intense rainfall. As observed in recent flood events, rainwater rapidly accumulates along roads, walkways, and building surroundings, disrupting mobility and campus operations. These incidents suggest that the flooding is not solely caused by prolonged rainfall, but possibly by a combination of high rainfall intensity, insufficient drainage capacity, surface runoff patterns, and campus topography.

Thus, the study seeks to address the primary problem:

Why does intense pluvial flooding occur within selected areas of the USTP-CDO campus, and how does rainfall intensity influence surface runoff behavior and flood accumulation?

Objectives of the Study:

General Objective:

- To develop a 3D digital twin-based system that uses rainfall intensity data from tipping rain gauges to simulate and analyze potential flood-prone conditions within the USTP-CDO campus.

Specific Objectives:

- To construct a 3D digital twin of the USTP-CDO campus, including drainage and map layout.
- To simulate rainfall-driven surface runoff based on collected rainfall data.
- To evaluate the system's reliability and usability in regards to simulating and analyzing flood-prone areas through 3D visualization.

Research Locale and Respondents:

The study will be conducted within selected flood-prone areas of the University of Science and Technology of Southern Philippines – Cagayan de Oro (USTP-CDO) campus. The focus will be on specific zones that historically experience surface water accumulation during heavy rainfall, including roadways, walkways, and areas near drainage systems. These locations will serve as the physical basis for the 2D digital twin model and rainfall-runoff simulation.

Theoretical Conceptual Framework:

The development of this Modular IoT-Based Digital Twin adopts the V-Model framework to ensure rigorous reliability through a systematic process of verification and validation. By pairing each design stage, such as the five layer system architecture including Input, Local Processing, Transmission, Backend, and Output, with a specific testing phase, the model ensures that critical hardware like the ESP32 microprocessors and tipping rain gauges are individually verified for accuracy before being integrated into the broader campus scale

simulation.

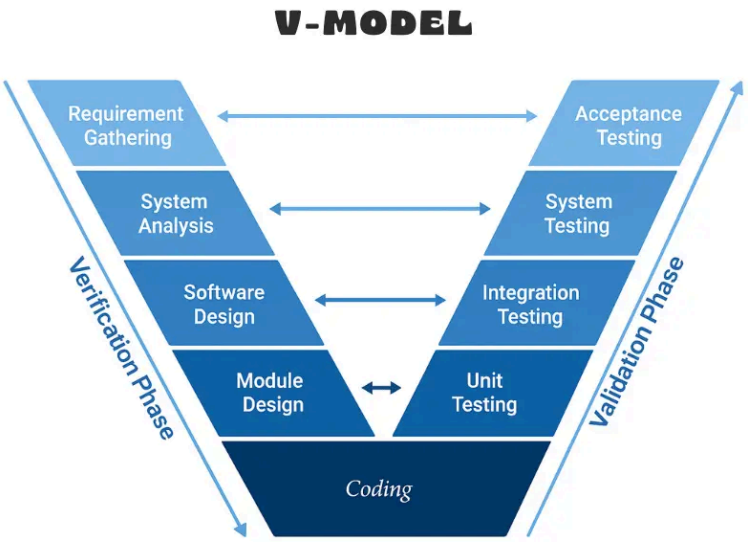

Figure 1. Theoretical Framework

Research Paradigm:

This section explains the research paradigm of this proposed study.

Input	Process	Output
• Rainfall intensity data from tipping rain gauge	• Data collection and logging of rainfall intensity	• 3D simulated surface runoff behavior
• Campus 3D surface and elevation layout	• Rainfall classification and 3D modeling of runoff	• Identification of flood-prone areas via 3D depth maps

Table 1. System Research Paradigm

Research Questions and Hypothesis:

- Why does intense pluvial flooding occur within selected areas of the USTP-CDO campus?
- What rainfall intensity levels contribute to surface water accumulation within the campus?
- How does the campus layout and drainage system influence flood behavior under varying rainfall conditions?

H_0 (Null Hypothesis):
Rainfall intensity has no significant influence on surface water accumulation and flood-prone area formation within the selected campus areas.

H_1 (Alternative Hypothesis):
Rainfall intensity significantly influences surface water accumulation and contributes to the formation of flood-prone areas within the selected campus areas.

Proposed Methodology:

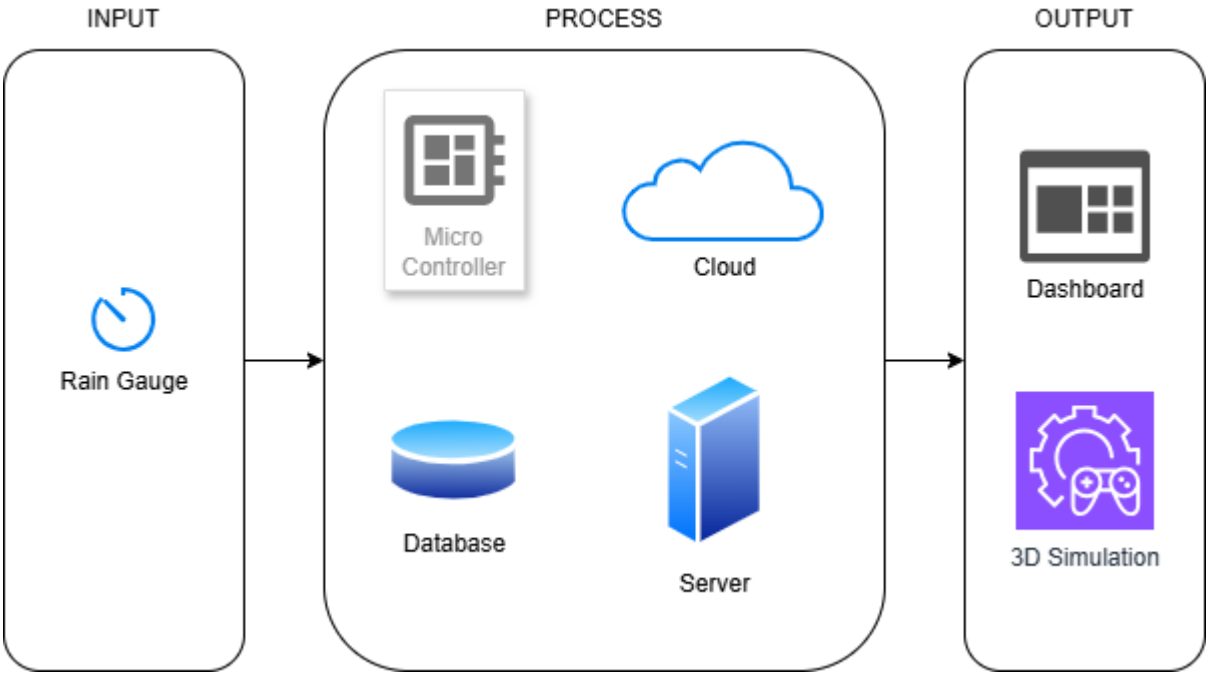

Figure 2: Conceptual Framework

The conceptual framework of this study is anchored on the Input-Process-Output (IPO) model, which facilitates the systematic transformation of localized environmental data into a functional 3D digital twin. This approach ensures that every physical measurement taken within the USTP-CDO campus is processed through a validated pipeline to yield accurate 3D flood-risk predictions.

- **Input Layer:** This stage involves the collection of rainfall intensity through a modular IoT sensing system featuring a singular tipping rain gauge sensor. It also integrates the geospatial constraints and topographical elevations of the campus, which serve as the foundation for the 3D virtual environment.

- **Process Layer:** The processing stage is governed by the V-Model methodology. It encompasses the real-time transmission of data via ESP32 microcontrollers to a cloud database, followed by analytical computations for surface runoff and volumetric accumulation. Crucially, this phase includes rigorous unit and integration testing to verify sensor accuracy and data integrity before the 3D simulation is generated.

- **Output Layer:** The final stage results in a 3D Digital Twin Dashboard hosted on a Raspberry Pi-powered kiosk. This interface provides real-time 3D visualizations of campus conditions, historical rainfall trends, and predictive flood-risk assessments, allowing for informed decision-making and improved institutional flood resilience.

Significance of the Study:

This study develops a campus-specific 2D digital twin driven by actual rainfall intensity data. This allows for a more accurate analysis of how rainfall translates into surface runoff and flood accumulation within USTP-CDO. Its significance is detailed below:

- **University Administration and Facilities Management:** The study provides a simulation-based tool that can support infrastructure planning, drainage improvement strategies, and flood mitigation decisions. By identifying flood-prone zones and rainfall thresholds that trigger flooding, the administration can prioritize upgrades and maintenance efforts more effectively.
- **University Students:** The study provides an easy-to-use tool that lets students know where it is most prone to pluvial flooding and is more susceptible to being stranded whenever strong and intense rainfall is present.
- **Future Researchers:** This study may serve as a baseline model that can be expanded to include real-time monitoring, multi-sensor integration, longer-term rainfall analysis, or even climate trend simulations. The modular structure of the proposed system allows scalability and further innovation.

Components and Cost:

The following materials will be used in the study.

Component	Price
Tipping Rain Gauge(x3)	₱ 8,100
Solar Panel(10w, 6v) (x3)	₱ 1,074
Battery (6000mAh)(x4)	₱ 1,200
Humidity & Temperature Sensor(x3)	₱ 690
Esp32 Microcontroller(x3)	₱ 180
Raspberry Pi	₱ 9,000
Touch Screen Monitor(10.1 inch)	₱ 4000
Total Cost:	**₱ 24,244**

Table 2. Hardware Components & Cost

Proposed Research Timeline:

Task	Duration

Drafting of Concept Paper	**9 weeks**
Development of Frameworks	**2 weeks**
Data Gathering and Documentation	**3 weeks**
Analysis and Synthesis of Findings	**6 weeks**
Testing and Revision of Prototypes	**3 weeks**
Final Concept Paper	**2 weeks**

Table 3. Proposed Research Timeline

References:

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